

CBSE Test Paper 03
Chapter 10 Mechanical Properties of Fluids

1. Density is defined as **1**
 - a. volume of 1 kg of the material
 - b. mass per unit volume
 - c. volume per unit mass
 - d. volume of 10 kg of the material
 2. The coefficient of viscosity for a fluid is defined as the ratio of **1**
 - a. shearing stress to the flow rate
 - b. shearing stress to the strain rate
 - c. compressive stress to the strain rate
 - d. tensile stress to the strain rate
 3. A man in a lift is holding a can filled with water but having a hole in its bottom. When will the drainage of water be maximum? **1**
 - a. when the lift goes down with uniform velocity
 - b. when the lift is stationary
 - c. when the lift is going down with acceleration
 - d. when the lift is going up with acceleration
 4. The height of mercury column in a simple barometer is h . As tube is inclined to the vertical at an angle α the length of the mercury column along the length of the tube is l , then **1**
 - a. $l = h/\cos\alpha$
 - b. $l = h \sin\alpha$
 - c. $l = h \cos\alpha$
 - d. $l = h$
 5. The surface tension of soap solution is $25 \times 10^{-3} \text{ Nm}^{-1}$. The excess pressure inside a soap bubble of diameter 1 cm is **1**
 - a. 15 Pa
 - b. 20 Pa
 - c. 5 Pa
 - d. 10 Pa
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6. Why is it difficult to stop bleeding from a cut in the human body at high altitudes? **1**
 7. The blood pressure of humans is greater at the feet than at the brain? **1**
 8. Is viscosity a vector? **1**
 9. A mercury barometer gives a reading of 74.5 cm. Calculate the atmospheric pressure at the place. Given that density of mercury = $13.6 \times 10^3 \text{ kg m}^{-3}$.
 10. Why is it painful to walk barefooted on a road covered with pebbles having sharp edges? **2**
 11. The terminal velocity of a tiny droplet is v . N number of such identical droplets combine together forming a bigger drop. Find the terminal velocity of the bigger drop. **2**
 12. Two mercury droplets of radii 0.1 cm. and 0.2 cm. collapse into one single drop. What amount of energy is released? The surface tension of mercury $T = 435.5 \times 10^{-3} \text{ N m}^{-1}$. **3**
 13. The flow rate of water is 0.58 L/min from a tap of diameter of 1.30 cm. After some time, the flow rate is increased to 4 L/min. Determine the nature of the flow for both the flow rates. The coefficient of viscosity of water is $10^{-3} \text{ Pa} \cdot \text{s}$ and the density of water is 10^3 kg/m^3 . **3**
 14. The free surface of oil in a tanker, at rest, is horizontal. If the tanker starts accelerating the free surface will be tilted by an angle θ . If the acceleration is $a \text{ ms}^{-2}$, what will be the slope of the free surface? **3**
 15. Two narrow bores of diameters 3.0 mm and 6.0 mm are joined together to form a U-tube open at both ends. If the U-tube contains water, what is the difference in its levels in the two limbs of the tube? Surface tension of water at the temperature of the experiment is $7.3 \times 10^{-2} \text{ N m}^{-1}$. Take the angle of contact to be zero and density of water to be $1.0 \times 10^3 \text{ kg m}^{-3}$ ($g = 9.8 \text{ m s}^{-2}$). **5**
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